```python
from sklearn.cluster import KMeans
import numpy as np

data = np.array([
    [1, 2],
    [2, 3],
    [3, 2],
    [8, 9],
    [9, 8],
    [10, 9]
])

model = KMeans(n_clusters=2, random_state=0, n_init=10)
model.fit(data)

print("PATTERN RECOGNITION")
print("---------------------------")
print("Input Data:")
print(data)

print("\nDetected Patterns:")
for i, point in enumerate(data):
    print(point, "belongs to Cluster", model.labels_[i])

print("\nCluster Centers:")
print(model.cluster_centers_)
print("\nNumber of Patterns Detected:", len(set(model.labels_)))
```

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/user/Downloads/AIES.py ==========
========
PATTERN RECOGNITION
---------------------------
Input Data:
[[ 1  2]
 [ 2  3]
 [ 3  2]
 [ 8  9]
 [ 9  8]
 [10  9]]

Detected Patterns:
[1 2] belongs to Cluster 1
[2 3] belongs to Cluster 1
[3 2] belongs to Cluster 1
[8 9] belongs to Cluster 0
[9 8] belongs to Cluster 0
[10  9] belongs to Cluster 0

Cluster Centers:
[[9.         8.66666667]
 [2.         2.33333333]]

Number of Patterns Detected: 2
>>>
```